Input: $x \in \{0,1\}^n$

Output: $a \in \{0,1\}$

Initialize $v \leftarrow$ root of T, $p \leftarrow ?^n$
while v is not a leaf **do**
    $(J, \ell) \leftarrow \mathsf{BuildList}(v, p)$
    Find the first element $i^* \in J$ with $x_{i^*} = 1$ or set $i^* = \perp$ if none exists
    FOUND $\leftarrow 0$
    for $j \in J$ **do**
        Sample $\boldsymbol{\eta} \sim \mathsf{Ber}(\delta_j/(2-\delta_j))$
        if $j = i^*$ or $\boldsymbol{\eta} = 1$ **then**
            $p_j \leftarrow \star$
            $u \leftarrow w_j$
            FOUND $\leftarrow 1$
            break
        end if
        $p_j \leftarrow 0$
    end for
    if FOUND $= 1$ **then**
        Query $\langle x, Q^u \rangle$ and set b^u as the outcome
        Move $v \leftarrow \mathrm{child}(u, b^u)$
    else
        Update $v \leftarrow \ell$
    end if
end while
return $L(v)$